

**Dataset:** <https://www.kaggle.com/competitions/agriyield-2025/data>

## **Which modeling approach performs best for yield prediction?**

Compare:

- Linear regression
- Random Forest
- Gradient Boosting (XGBoost/lightGBM)
- Shallow neural networks

## **Introduction**

When considering the future effects of climate change on global populations, agriculture is a unifying concern. Every country, regardless of economic development, industrialization, population density, or political power, will need to consider the changing patterns of climate change if they hope to maintain the harvests needed to achieve national food security.

Climatic and surface conditions are significant determinants of agricultural productivity, harvest efficiency, and crop quality. Every crop requires a unique range of necessary environmental and mechanical inputs in order to achieve maximum efficiency during growth, harvest, and reproduction. These elements (like daily temperature and precipitation, soil quality and content, and slope and elevation characteristics) are vulnerable to changes in regional climate and global warming, which can disrupt the ideal conditions for crop production.

Changes to weather patterns and landscape characteristics can happen swiftly and without warning, taking farmers by surprise and impeding critical harvest yields. There has been significant research concerning the prediction of agricultural yields including Haggerty et al. (2025), which uses climate indicators to rank the suitability of landscapes in southern Brazil for corn growth and production. Without knowledge of feature importance, however, these models would produce insignificant valuations of crop predictions and remain unhelpful for agricultural policy-making, food security evaluations, and resource allocation.

The data we selected from Kaggle explores maize (corn) yields as a target of environmental features like soil characteristics, weather, and remote sensing properties. The challenge of this Kaggle competition was to “support precision agriculture by enabling early yield forecasting and smarter farm management”. By using this dataset, we hope to identify which method of machine learning and data analysis from our class is the most applicable to this kind of agricultural prediction and explore the feature importance of environmental indicators for yield predictions. Detailed knowledge on modelling methods and feature importance strengthens modelling outputs and better informs both agriculture producers and consumers.

The data we received from Kaggle contains three files: the training dataset with input features and yield values, the test dataset with features only (for prediction), and a sample submission template. The training dataset contains the following information:

Features:

- field\_id: Unique identifier for each field
- soil\_ph: Soil acidity (pH scale)
- organic\_matter: Percentage of organic matter in the soil
- sand\_pct: Percentage of sand in soil composition
- temperature: Average temperature (°C) during the growing season
- humidity: Average relative humidity (%) during growing season
- rainfall: Total rainfall (mm) during growing season
- ndvi: Normalized Difference Vegetation Index (remote sensing)

Target:

- yield: Maize yield measured in kilograms per hectare (kg/ha)

## Model Selection

To establish a metric with which we could evaluate a model's predictive power, we calculated the mean RMSE as the baseline for the five models. This value was 435.33, and of all the baseline models only Linear Regression outperformed it with a value of 435.25. Given that Linear Regression is unique in being the only linear model in our experiment, we decided to also explore the best non-linear model (Gradient Boosting) to ensure that we capture any non-linear structures in the data. The closeness in values between the Linear Regression RMSE and the predicted mean told us that we would likely not find much non-linearity, but we decided to err on the side of caution given the minimal contextual information provided with the dataset. After multiple rounds of hyperparameter tuning, the performance of the Gradient Boost model improved from 436.16 to 435.23 with the following results:

**Best non-linear model (from baseline): Gradient Boosting**

**Running hyperparameter tuning for: Gradient Boosting**  
**Fitting 3 folds for each of 15 candidates, totalling 45 fits**

**Best hyperparameters: {'subsample': 0.7, 'n\_estimators': 300, 'max\_depth': 3, 'learning\_rate': 0.01}**

**Validation RMSE (tuned Gradient Boosting): 435.2348**

This improvement, and the fact that Gradient Boosting has outperformed both the Linear Regression model and the predicted mean, made us feel confident in selecting it for the feature importance analysis.

## Feature Importance Analysis

The tuned Gradient Boosting model provides a non-linear perspective on how the environmental and soil variables contribute to maize yield prediction. Although the overall predictive power remains limited, it provides insights that form a complete picture of the effects of climate change on environmental variables. Temperature emerged as the most influential variable (importance = 0.212). This aligns with established agronomic understanding: maize is sensitive to thermal conditions throughout the growing season, and slight fluctuations can affect development rates, biomass accumulation, and reproductive success. Even modest temperature differences may translate into small yield variations detectable by tree-based models.

Rainfall was the second most influential factor (importance = 0.210), consistent with the strong dependence of maize on adequate water supply. Total precipitation affects soil moisture levels, stress during silking and grain filling, and overall crop vigor. Humidity (importance = 0.153) also played a moderate role, which is plausible given its indirect effects on transpiration, evaporative demand, and potential disease pressure. Sand percentage (0.148) contributed meaningfully as well; soils high in sand typically drain quickly and retain fewer nutrients, making crops more susceptible to drought stress and nutrient deficiencies. These interpretations demonstrate that Gradient Boosting captures agronomically reasonable patterns even when relationships are weak.

The remaining features—NDVI (importance = 0.106), organic matter (0.098), and soil pH (0.072)—were less influential. The modest importance assigned to NDVI may be due to limited temporal variability or the use of a single-season snapshot rather than a time-series representation of vegetation health. Organic matter and pH, while important soil quality indicators in many cropping systems, may not vary sufficiently across fields in the dataset to produce strong predictive effects. Overall, the Gradient Boosting and Linear Regression results suggest that climate-related factors (temperature, rainfall, humidity) and soil physical structure (sand content) have comparatively greater influence on yield predictions.

## Conclusion

Using the results from our feature analysis and model exploration, we can tell that, even with lower-quality data, corn yield is highly impacted by precipitation and moisture. Temperature is an obvious contender for importance in predicting crop growth, as daily temperatures directly correlate to Growing Degree Days (or GDDs), which farmers use to estimate growing windows and harvest times. Precipitation and humidity, ranking second and third-highest in importance, are intertwined variables, as atmospheric moisture is a controlling factor for rainfall. Minimum rainfall allowances are a necessity for corn growth, even on farms with access to irrigation. Similarly, the final important feature, soil sand percentage, is likely an influential feature because of its relation to soil porosity, permeability, and water content.

The emergence of these variables as the most important features also highlights the impact of moisture on crop yield overall. Any variations in moisture, whether through temperature or water content, have the highest potential to disrupt crop yields. Therefore, the direct relationship between moisture and climate change means that climate change cannot be ignored in discussions of food security and agricultural preparedness. Soil organic matter content, soil pH, and NDVI values describe the physical and chemical properties of these maize farms, and are evidently less dependable indicators of crop growth.